Artificial Intelligence Foundations (CIS4049-N)

Week 1 – What's AI – Part 3

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A short History of AI

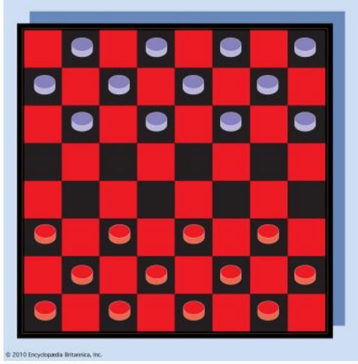
Birth of AI

Teesside University

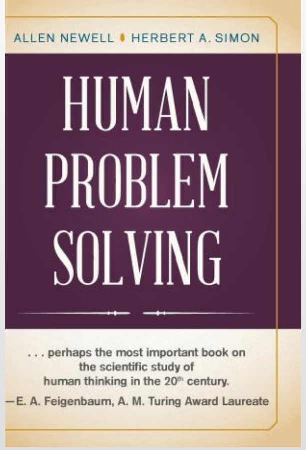
- **1956**: Workshop at Dartmouth College; attendees: John McCarthy, Marvin Minsky, Claude Shannon, etc.

- **Aim** for general principles: Every aspect of **learning** or any other feature of **intelligence** can be so precisely described that a **machine** can be made to simulate it.

Birth of AI



- **1952 - Checkers**: Samuel's program learned weights and played at strong amateur level



- **1955 - Problem Solving**: Newell & Simon's Logic Theorist: prove theorems in Principia Mathematica using search and heuristics; later, General Problem Solver (GPS)

Overwhelming Optimism

- *“Machines will be capable, within twenty years, of doing any work a man can do.”*

  — Herbert Simon

- *“Within 10 years the problems of artificial intelligence will be substantially solved.”*

  — Marvin Minsky

- *“I visualize a time when we will be to robots what dogs are to humans, and I’m rooting for the machines. “*

  — Claude Shannon

.. underwhelming Results

- Example: **machine translation.**

"The spirit is willing, but the flesh is weak". (Russian)

↓

"The vodka is good, but the meat is rotten".

- **1966**: ALPAC report cut off government funding for MT - **First "AI winter"**

Implications of early era

- **Problems**:
  - Limited computation: search space grew exponentially, outpacing hardware (100! ≈ 10157 > 1080)
  - Limited information: complexity of AI problems (number of words, objects, concepts in the world)
- **Contributions**:
  - **Lisp** (1958), garbage collection, time-sharing (John McCarthy)
  - **Key paradigm**: separate **modeling** and **inference**

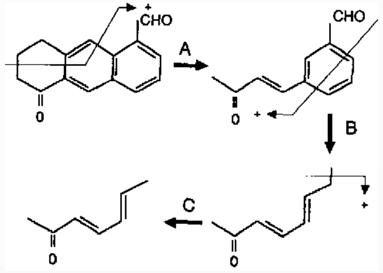
Knowledge-based Systems

- **70-80s: Expert Systems -** elicit specific domain knowledge from experts in form of rules:

if [premises] then [conclusion]

Knowledge-based Systems

- **DENDRAL**: infer molecular structure from mass spectrometry



- **MYCIN**: diagnose blood infections, recommend antibiotics

- **XCON**: convert customer orders into parts specification;
save DEC $40 million a year by 1986

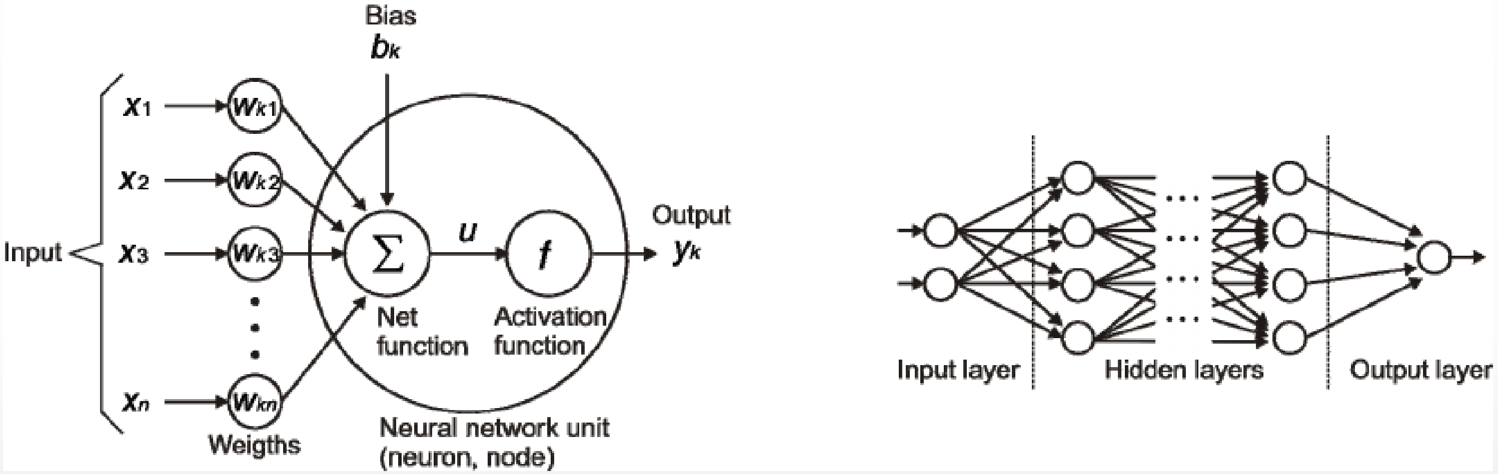
Knowledge-based Systems

- **Contributions**:
  - First **real application** that impacted industry
  - Knowledge helped curb the exponential growth
- **Problems**:
  - Knowledge is not deterministic rules, need to model uncertainty
  - Requires considerable manual effort to create rules, hard to maintain
- **1987**: Collapse of Lisp machines - **second AI winter**

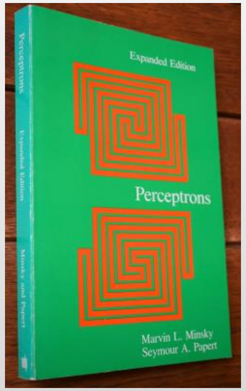
Artificial Neural Networks

Teesside University

- **1943: artificial neural networks**, connect neural circuitry and logic (McCulloch/Pitts)

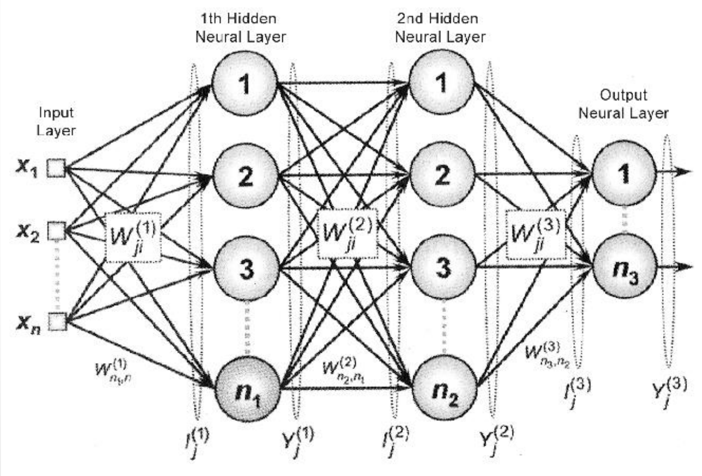


- **1969: Perceptrons** book showed that linear models could not solve XOR, killed neural nets research (Minsky/Papert)



Training Networks

- **1986:** popularization of backpropagation for training multi-layer networks (Rumelhardt, Hinton, Williams)



- **1989:** applied convolutional neural networks to recognizing handwritten digits for USPS (LeCun)

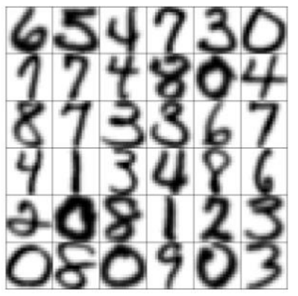
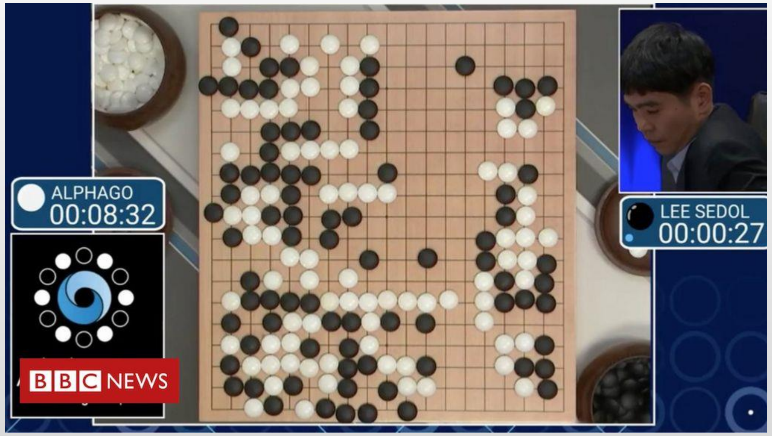


Figure : USPS data

Deep Learning

- **Alex Net (2012):** huge gains in object recognition; transformed computer vision community overnight



- **AlphaGo (2016):** deep reinforcement learning, defeat world champion Lee Sedol